



Lecture 4: Primitive Recursive Functions, Composition, μ -Recursion

Primitive Functions:

- **Null function:** $0: \mathbb{N}_0 \rightarrow \mathbb{N}_0 \quad 0(x) = 0$
- **Successor function:** $S: \mathbb{N}_0 \rightarrow \mathbb{N}_0 \quad S(x) = x + 1$
- **Projection:** $\pi_i^k: \mathbb{N}_0^k \rightarrow \mathbb{N}_0 \quad \pi_i^k(x_1, \dots, x_k) = x_i$

Defining Primitive Recursive Functions (PRFs):

Let $g: \mathbb{N}_0^n \rightarrow \mathbb{N}_0$, $h: \mathbb{N}_0^{n+2} \rightarrow \mathbb{N}_0$ be primitive recursive. Then $f: \mathbb{N}_0^{n+1} \rightarrow \mathbb{N}_0$ with

$$f(x_1, \dots, x_n, 0) = g(x_1, \dots, x_n)$$

$$f(x_1, \dots, x_n, S(y)) = h(x_1, \dots, x_n, y, f(x_1, \dots, x_n, y))$$

is also primitive recursive.

PRF #1: Addition

Definition:

$$add(x, 0) = x$$

$$add(x, S(y)) = S(\pi_3^3(x, y, add(x, y)))$$

Example:

$$\begin{aligned} add(2,3) &= S(\pi_3^3(2,3, add(2,2))) \\ &= S\left(\pi_3^3\left(2,3, S\left(\pi_3^3(2,3, add(2,1))\right)\right)\right) \\ &= S\left(\pi_3^3\left(2,3, S\left(\pi_3^3\left(2,3, S\left(\pi_3^3(2,3, add(2,0))\right)\right)\right)\right)\right) \\ &= S\left(\pi_3^3\left(2,3, S\left(\pi_3^3\left(2,3, S(\pi_3^3(2,3,2))\right)\right)\right)\right) \\ &= S\left(\pi_3^3\left(2,3, S\left(\pi_3^3(2,3, S(2))\right)\right)\right) \\ &= S\left(\pi_3^3\left(2,3, S(\pi_3^3(2,3,3))\right)\right) \end{aligned}$$



$$\begin{aligned}
 &= S(\pi_3^3(2, 3, S(3))) \\
 &= S(\pi_3^3(2, 3, 4)) \\
 &= S(4) \\
 &= 5
 \end{aligned}$$

Composition of Primitive Recursive Functions:

Let $h_1, \dots, h_n: \mathbb{N}_0 \rightarrow \mathbb{N}_0$, $g: \mathbb{N}_0^n \rightarrow \mathbb{N}_0$ be primitive recursive, then $f: \mathbb{N}_0^k \rightarrow \mathbb{N}_0$ with

$$f(x_1, \dots, x_k) = g(h_1(x_1, \dots, x_k), h_2(x_1, \dots, x_k), \dots, h_n(x_1, \dots, x_k))$$

is also primitive recursive.

or

$h_1, \dots, h_n: \mathbb{N}_0 \rightarrow \mathbb{N}_0$, $g: \mathbb{N}_0^n \rightarrow \mathbb{N}_0$ is a primitive rekursive function, then $f: \mathbb{N}_0^k \rightarrow \mathbb{N}_0$ with

$$f(x_1, \dots, x_k) = g \circ (h_1, h_2, \dots, h_n)(x_1, \dots, x_n)$$

is also primitive recursive.

PRF #2: Multiplication

Definition:

$$mult(x, 0) = 0$$

$$mult(x, S(y)) = add \circ (\pi_1^3, \pi_3^3)(x, y, mult(x, y))$$

Example:

$$\begin{aligned}
 mult(3, 2) &= add \circ (\pi_1^3, \pi_3^3)(3, 2, mult(3, 1)) \\
 &= add \circ (\pi_1^3, \pi_3^3)(3, 2, add \circ (\pi_1^3, \pi_3^3)(3, 2, mult(3, 0))) \\
 &= add \circ (\pi_1^3, \pi_3^3)(3, 2, add \circ (\pi_1^3, \pi_3^3)(3, 2, 0)) \\
 &= add \circ (\pi_1^3, \pi_3^3)(3, 2, add(3, 0)) \\
 &= add \circ (\pi_1^3, \pi_3^3)(3, 2, 3) \\
 &= add(3, 3) \\
 &= 6
 \end{aligned}$$



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PRF #3: Subtraction

$$sub: N_0 \times N_0 \rightarrow N_0 \quad sub(x, y) = \begin{cases} x - y, & x \geq y \\ 0, & x < y \end{cases}$$

To show that the sub function is primitive recursive, we need a predecessor function P . The predecessor function P acts as the opposite of the successor function S and is recursively defined by the rules:

$$P: N_0 \rightarrow N_0 \quad P(x) = \begin{cases} x - 1, & x \geq 1 \\ 0, & x = 0 \end{cases}$$

Definition:

$$P(0) = 0 \\ P(S(x)) = \pi_1^2(x, P(x))$$

Example:

Predecessor of 4?

$$\begin{aligned} P(S(3)) &= \pi_1^2(3, P(2)) \\ &= \pi_1^2(3, \pi_1^2(3, P(1))) \\ &= \pi_1^2(3, \pi_1^2(3, \pi_1^2(3, P(0)))) \\ &= \pi_1^2(3, \pi_1^2(3, \pi_1^2(3, 0))) \\ &= \pi_1^2(3, \pi_1^2(3, 3)) \\ &= \pi_1^2(3, 3) \\ &= 3 \end{aligned}$$

With the help of the predecessor function P , now we can show that the sub function is primitive recursive:

Definition:

$$sub(x, 0) = x \\ sub(x, S(y)) = P(\pi_3^3(x, y, sub(x, y)))$$

Example:

$$\begin{aligned} sub(4, 2) &= P(\pi_3^3(4, 2, sub(4, 1))) \\ &= P(\pi_3^3(4, 2, P(\pi_3^3(4, 2, sub(4, 0))))) \\ &= P(\pi_3^3(4, 2, P(\pi_3^3(4, 2, 4)))) \end{aligned}$$



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$$\begin{aligned}
 &= P\left(\pi_3^3(4,2,P(4))\right) \\
 &= P\left(\pi_3^3(4,2,3)\right) \\
 &= P(3) \\
 &= 2
 \end{aligned}$$

PRF #4: Exponent

$$exp: N_0 \times N_0 \rightarrow N_0 \quad exp(b, y) = b^y$$

Definition:

$$\begin{aligned}
 exp(x, 0) &= 1 \\
 exp(x, S(y)) &= mult \circ (\pi_1^3, \pi_3^3)(x, y, exp(x, y))
 \end{aligned}$$

Example:

$$\begin{aligned}
 exp(2,3) &= mult \circ (\pi_1^3, \pi_3^3)(2,3, exp(2,2)) \\
 &= mult \circ (\pi_1^3, \pi_3^3)\left(2,3, mult \circ (\pi_1^3, \pi_3^3)(2,3, exp(2,1))\right) \\
 &= mult \circ (\pi_1^3, \pi_3^3)\left(2,3, mult \circ (\pi_1^3, \pi_3^3)\left(2,3, mult \circ (\pi_1^3, \pi_3^3)(2,3, exp(2,0))\right)\right) \\
 &= mult \circ (\pi_1^3, \pi_3^3)\left(2,3, mult \circ (\pi_1^3, \pi_3^3)(2,3, mult \circ (\pi_1^3, \pi_3^3)(2,3,1))\right) \\
 &= mult \circ (\pi_1^3, \pi_3^3)\left(2,3, mult \circ (\pi_1^3, \pi_3^3)(2,3, mult(2,1))\right) \\
 &= mult \circ (\pi_1^3, \pi_3^3)\left(2,3, mult \circ (\pi_1^3, \pi_3^3)(2,3,2)\right) \\
 &= mult \circ (\pi_1^3, \pi_3^3)(2,3, mult(2,2)) \\
 &= mult \circ (\pi_1^3, \pi_3^3)(2,3,4) \\
 &= mult(2,4) \\
 &= 8
 \end{aligned}$$

Homework for next week:

$$equal: N_0 \times N_0 \rightarrow N_0 \quad equal(x, y) = \begin{cases} 1, & x = y \\ 0 & x \neq y \end{cases}$$



Primitive Recursive Functions (PRF) = Loop-Computable Functions

μ -Operator (Mikro-Operator):

$$f: \mathbb{N}_0^{n+1} \rightarrow \mathbb{N}_0 \quad \mu[f]: \mathbb{N}_0^n \rightarrow \mathbb{N}_0$$

$$\mu[f](x_1, \dots, x_n) = \min\{z \mid (x_1, \dots, x_n, y) \in \text{Def}(f) \text{ for } y \leq z \text{ and } f(x_1, \dots, x_n, z) = 0\}$$

Example:

$$\log: \mathbb{N}_0^2 \rightarrow \mathbb{N}_0 \quad \log_b(x, y) = x - \exp(b, y)$$

Assumption: b is constant. The μ -recursive computation of $\log: \mathbb{N}_0^2 \rightarrow \mathbb{N}_0$ with the following definition:

$$\mu[\log](x) = \min\{z \mid \log(x, y) \in \text{Def}(f) \text{ for } y \leq z \text{ and } \log(x, z) = 0\}$$

The procedure for $b = 2, x = 4$

$$\mu[\log](4, 0) = 3$$

$$\mu[\log](4, 1) = 2$$

$$\mu[\log](4, 2) = 0 \leftarrow \text{The correct result (Note: The log function is not total)}$$

$\Rightarrow$ The class that comes together by combining/nesting **primitive recursion**, **composition**, and/or the **μ -operator**, is called as the **class of μ -recursive functions** $\mathbb{P}$ (also referred to as the **class of partial recursive functions**).

$\Rightarrow$ A distinction is made between the set of μ -recursive functions that are **total**, i.e. $\mathbb{R}$.

Computability concepts: TM, TA, QA, 2PDAs, loops, while, goto, class of μ -recursive functions

Ackermann Function

From last week we know that

$\Rightarrow$ f : loop-computable functions $\rightarrow$ total computable functions

Question: How about the other direction?

$\Rightarrow$ loop-computable functions $\leftarrow$ total computable functions ?

$\notin$
Ackermann function



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Ackermann function, referred to as $ack(n, m)$ or $ack_n(m)$, is defined as follows:

$$\begin{aligned}ack(0, m) &= m + 1 \\ack(n, 0) &= ack(n - 1, 1) \\ack(n, m) &= ack(n - 1, ack(n, m - 1))\end{aligned}$$

Example #1:

$$\begin{aligned}ack(1, 2) &= ack(0, ack(1, 1)) \\&= ack(0, ack(0, ack(1, 0))) \\&= ack(0, ack(0, ack(0, 1))) \\&= ack(0, ack(0, 2)) \\&= ack(0, 3) \\&= 4\end{aligned}$$

Example #2:

$$\begin{aligned}ack(2, 2) &= ack(1, ack(2, 1)) \\&= ack(1, ack(1, ack(2, 0))) \\&= ack(1, ack(1, ack(1, 1))) \\&= ack(1, ack(1, ack(0, ack(1, 0)))) \\&= ack(1, ack(1, ack(0, ack(0, 1)))) \\&= ack(1, ack(1, ack(0, 2))) \\&= ack(1, ack(1, 3)) \\&= ack(1, ack(0, ack(1, 2))) \\&= ack(1, ack(0, ack(0, ack(1, 1)))) \\&= ack(1, ack(0, ack(0, ack(0, ack(1, 0)))) \\&= ack(1, ack(0, ack(0, ack(0, ack(0, 1)))) \\&= ack(1, ack(0, ack(0, ack(0, 2)))) \\&= ack(1, ack(0, ack(0, 3))) \\&= ack(1, ack(0, ack(0, 3))) \\&= ack(1, ack(0, 4)) \\&= ack(1, 5) \\&= ack(0, ack(1, 4)) \\&= ack(0, ack(0, ack(1, 3)))\end{aligned}$$



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$$\begin{aligned}
 &= \text{ack}(0, \text{ack}(0, \text{ack}(0, \text{ack}(1, 2)))) \\
 &= \text{ack}(0, \text{ack}(0, \text{ack}(0, \text{ack}(0, \text{ack}(1, 1))))) \\
 &= \text{ack}(0, \text{ack}(0, \text{ack}(0, \text{ack}(0, \text{ack}(0, \text{ack}(1, 0))))) \\
 &= \text{ack}(0, \text{ack}(0, \text{ack}(0, \text{ack}(0, \text{ack}(0, \text{ack}(0, 1))))) \\
 &= \text{ack}(0, \text{ack}(0, \text{ack}(0, \text{ack}(0, \text{ack}(0, \text{ack}(0, 2))))) \\
 &= \text{ack}(0, \text{ack}(0, \text{ack}(0, \text{ack}(0, \text{ack}(0, 3))))) \\
 &= \text{ack}(0, \text{ack}(0, \text{ack}(0, 4))) \\
 &= \text{ack}(0, \text{ack}(0, 5)) \\
 &= \text{ack}(0, 6) \\
 &= 7
 \end{aligned}$$

Homework for next week:

$$\text{ack}(3, 3) = ?$$

Properties of Ackermann function:

$$\text{ack}(m, n)$$

$\Downarrow$

$\text{ack}_m(n) \rightarrow$ Family of Ackermann functions with $m=0, m=1, m=2$, etc.

$\Rightarrow$ It applies to all primitive recursive functions: In this sequence of Ackermann functions, there is a function that grows faster than the primitive recursive function being considered.

$$1. \text{ack}_i(x) \leq \text{ack}_{i+j}(x + y), \quad j, y \geq 0$$

$$2. \text{ack}_i(x + 1) \leq \text{ack}_{i+1}(x), \quad x \geq x_0$$

$$\text{ack}_i(x + j) \leq \text{ack}_{i+j}(x), \quad x \geq x_0 \quad \leftarrow \text{Generalization of the 2. expression above}$$

$$3. \forall f \in \text{loop} \exists i, x_0: f(x) \leq \text{ack}_i(x), \quad x \geq x_0$$